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Dept. of Electrical and Computer Engineering

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Preliminary Design

Project name - Geometric shaping in the FSO turbulent channel

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Students (name & ID):	Dolev Ishay 211721451 Adi Shlomo 211619226
Supervisors:	Prof Derevyanko Stanislav
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Geometric shaping in the FSO turbulent channel

Students Names: Ishay Dolev, Shlomo Adi

dolevish@post.bgu.ac.il

Adviser name: Prof Derevyanko Stanislav

Background: Free-space optical (FSO) communication enables high data rates but suffers from random attenuation due to atmospheric turbulence and additive white Gaussian noise (AWGN). Under weak turbulence, the intensity fading is well-modeled by a log-normal distribution.

Purpose: The project aims to improve the theoretical channel capacity, expressed as the Average Mutual Information (AMI), for an FSO system operating under weak turbulence and AWGN, by optimizing the structure of the transmitted constellation.

Objectives: To find the optimal locations of M points in a one-dimensional intensity modulation constellation that maximize the AMI for the given channel model, subject to a fixed average transmit power constraint, and to achieve a Shaping Gain of 0.2 dB to 1.0 dB compared to standard Uniform-PAM.

The innovation: Applying the Geometric Shaping (GS) technique, proven effective in other channels, specifically to the FSO channel model with log-normal turbulence and AWGN, using the Simulated Annealing (SA) optimization algorithm.

Proposed method: We use geometric shaping (GS) for a positive M-PAM constellation, optimizing the symbol locations to maximize AMI under a fixed average-power constraint, via simulated annealing (SA). Within each SA iteration, AMI is estimated by nested Monte Carlo with Gauss–Hermite quadrature (MC+GH) for efficient evaluation of the conditional likelihood, and validated by Monte Carlo with tabulated/interpolated likelihood on a nonuniform y -grid (MC-via-grid).

Expected results: We expect the optimized GS constellation to outperform the standard PAM constellation in terms of AMI for given average power constraint and SNR, thereby demonstrating the potential of geometric shaping for performance improvement in log-normal fading channels.

Key words: FSO, Geometric Shaping, Constellation Optimization, Log-Normal Fading, AMI, Simulated Annealing, Intensity Modulation, AWGN, Channel Capacity, Turbulence.

עיצוב גאומטרי בערוץ טורבולנטי בתקשורת אופטית במרחב חופשי

שמות הסטודנטים: ישי דולב, שלמה עדי

שם המנחה: פרופ' דרביאנקו סטניסלב

dolevish@post.bgu.ac.il

רקע: תקשורת אופטית בחלל חופשי (FSO) מאפשרת קצבי נתונים גבוהים אך סובלת מהנחתות אקראיות עקב טורבולנציה אטמוספירית ורעש גאומטרי אדיטיבי (AWGN). במצב של טורבולנציה חלשה, הנחתת העוצמה מתוארת היטב על ידי התפלגות לוג-נורמלית.

מטרה: הפריקט נועד לשפר את קיבולת הערוץ התאורטית, המבוססת על ידי האינפורמציה ההדדית הממוצעת (AMI), במערכת FSO הפועלת בתנאי טורבולנציה חלשה ורעש AWGN, באמצעות אופטימיזציה של מבנה הקונסטלציה המשודרת.

יעדים: למצוא את המיקומים האופטימליים של M נקודות בקונסטלציית אפנון עוצמה חד-ממדית שממקסמים את ה-AMI, בערוץ הנתון, תחת אילוץ של הספק שידור ממוצע קבוע, ולהשיג רווח עיצוב (Shaping Gain) של 0.2 dB עד 1.0 dB ביחס לקונסטלציית PAM סטנדרטית.

החידוש: יישום שיטת העיצוב הגאומטרי (GS), שהוכחה כיעילה בערוצים אחרים, ספציפית למודל הערוץ של FSO עם טורבולנציה לוג-נורמלית ורעש AWGN, תוך שימוש באלגוריתם אופטימיזציה מסוג Simulated Annealing (SA).

השיטה המוצעת: נשתמש בעיצוב גאומטרי (GS) לקונסטלציית M-PAM חיובי, כאשר מיקומי הסמלים מותאמים כדי למקסם AMI תחת אילוץ עוצמה ממוצעת קבוע, באמצעות SA. בכל איטרציית SA נחשב את ה-AMI בעזרת מונטה-קרלו מקונן עם קוודרטור גאוס-הרמיט (MC+GH) לחישוב יעיל של הנראות המותנית (conditional likelihood), ויאומת באמצעות מונטה קרלו (Monte Carlo) עם נראות (likelihood) מטבלה \מאינטרפולציה (tabulated/interpolated) על גבי רשת y לא-אחידה (MC-via-grid).

תוצאות צפויות: אנו צופים כי הקונסטלציה המעוצבת (GS) תשיג ערך AMI הגבוה מזה של קונסטלציית PAM סטנדרטית עבור הספק שידור ו-SNR נתון, ובכך תדגים את הפוטנציאל של עיצוב גאומטרי לשיפור ביצועים בערוצי דעיכה לוג-נורמלית.

מילות מפתח: תקשורת אופטית בחלל חופשי (FSO), עיצוב גאומטרי (GS), אופטימיזציית קונסטלציה, דעיכות לוג-נורמליות, אינפורמציה הדדית ממוצעת (AMI), הרפיה מדומה (SA), אפנון עוצמה, רעש גאומטרי לובן אדיטיבי (AWGN), קיבולת ערוץ, טורבולנציה.

Project goal

The primary goal of the project is to maximize the channel capacity of Free-Space Optical (FSO) communication systems operating under weak atmospheric turbulence (log-normal fading) and Additive White Gaussian Noise (AWGN), subject to an average transmit power constraint at a given SNR.

The project addresses the inherent inefficiency of using standard uniformly spaced constellations (Uniform-PAM) in these non-Gaussian channels, which leads to capacity loss. Consequently, the objective is to determine the optimal locations of the constellation points (Geometric Shaping) to overcome this inefficiency.

The project success metric is defined as achieving an Average Mutual Information (AMI) value strictly higher than that of a standard PAM constellation, reflecting a measurable Shaping Gain (e.g., > 0.2 dB).

Research Proposal

Geometric shaping in the FSO turbulent channel

Research Topic: The research topic is the optimization of constellation geometry for Intensity Modulation/Direct Detection (IM/DD) communication systems operating over Free-Space Optical (FSO) channels impaired by weak atmospheric turbulence and Additive White Gaussian Noise (AWGN). Specifically, we aim to maximize the Average Mutual Information (AMI), a measure of channel capacity, by strategically positioning the constellation points (Geometric Shaping) under an average power and SNR constraint.

Technology: The project utilizes Free-Space Optical (FSO) communication technology, which employs intensity-modulated laser beams for point-to-point atmospheric data transmission. The specific system configuration is based on Intensity Modulation/Direct Detection (IM/DD), where data is encoded onto the optical power and recovered using a photodetector.

Mathematical Research Model:

Channel Input/Output:

The input is a transmitted optical intensity x chosen from a positive, discrete M-ary constellation $\mathcal{X} = \{x_1, \dots, x_M\}$ with uniform probability $P(x_i) = 1/M$.

The output is the received electrical signal y , which is a continuous real value.

Channel Equation:

The relationship between input and output is given by $y = hRx + n$. Where:

x is the transmitted intensity ($x > 0$).

h is the random channel state (fading).

R is the detector's responsivity.

n is additive white Gaussian noise $n \sim \mathcal{N}(0, \sigma^2)$.

Component Distributions:

Channel Fading h : A positive random variable representing intensity attenuation due to weak turbulence. It follows a log-normal distribution, meaning its logarithm

$$t = \ln(h) \sim \mathcal{N}(\mu_t, \sigma_t^2) \text{ where } P_h(h) = \frac{1}{\sqrt{2\pi h \sigma_t}} \exp\left(-\frac{(\ln h - \mu_t)^2}{2\sigma_t^2}\right) \text{ for } h > 0$$

Noise n : Additive White Gaussian Noise $n \sim \mathcal{N}(0, \sigma_n^2)$ representing noise at the receiver.

Conditional PDF: The probability density of receiving y given x was transmitted is found by averaging the Gaussian noise effect over all possible fading states:

$$P(y|x) = \int_0^\infty P(y|x, h)P_h(h)dh.$$

Where $P(y|x, h) = \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(y-hRx)^2}{2\sigma_n^2}\right)$ is the Gaussian PDF conditioned on a specific fading state h .

Constraint: The average transmitted power is constrained: $P_{avg} = \frac{1}{M} \sum_{i=1}^M x_i^2$

Performance Metric (AMI): The metric to be maximized is the Average Mutual Information, calculated under the assumption of uniform inputs: $AMI = I(X; Y) = \sum_{i=1}^M P(x_i) \int_{-\infty}^{\infty} P(y|x_i) \log_2 \frac{P(y|x_i)}{P(y)} dy$. Where $P(y) = \sum_{j=1}^M P(x_j)P(y|x_j)$ is the PDF of the output signal. AMI represents the maximum reliably transmissible information rate for the given constellation and input distribution.

Research and Work Stages:

Literature Review: Deepen understanding of the log-normal channel model, AMI calculation methods (MC+GH, MC-via-grid), and the SA algorithm and its parameterization.

Implement Optimization Algorithm: Develop the SA framework, including proposing new constellation points, enforcing the power constraint, and implementing the Metropolis acceptance criterion. As a preliminary validation step, we have verified our SA implementation by comparing its results against published findings from reference [3], which utilizes a similar approach for constellation design. The results of this verification are presented in Appendix A.

Implement Base Functions: Develop and verify functions for calculating $P(y|x)$ and AMI using both MC+GH and MC-via-grid methods.

Integration: Combine the AMI calculation function (MC+GH) as the objective within the SA loop.

Run Optimization Simulations: Execute the SA algorithm for various channel parameters (constellation size M , average power P_{avg} , turbulence strength σ_X^2 , noise variance σ_n^2) to find optimal GS constellations.

Validation and Analysis: Verify calculated AMI values using an independent method (MC-via-grid). Quantify the AMI gain of optimized GS constellations compared to standard PAM. Analyze how this gain varies with different channel conditions and constellation sizes to evaluate the method's effectiveness across scenarios.

Potential Applications of Research Outcomes:

The primary practical outcomes will be optimized M-PAM constellation coordinates for log-normal FSO channels, offering quantifiable AMI gains over standard PAM. This enables designing FSO systems with potentially higher rates, better power efficiency, or extended reach. Additionally, the developed optimization methods provide a valuable framework adaptable for similar research problems.

System Schema:

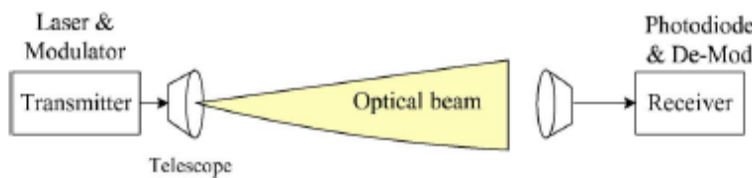


Figure 1: Block diagram of an FSO link [1].

Literature review

FSO Channel Characteristics and Statistical Modeling

Free-space optical (FSO) communication offers high data rates but is inherently subject to stochastic atmospheric impairments. Unlike deterministic channels, the received signal intensity in FSO fluctuates randomly due to atmospheric effects and pointing/alignment-related impairments. Farid and Hranilovic [1] emphasize the need for capacity-oriented analysis of FSO links under such random impairments. Following their established modeling approach [1], we model the intensity fluctuations in the weak turbulence regime using a log-normal distribution. As noted in [1], this model has been extensively validated through experimental measurements for these specific atmospheric conditions.

System Constraints and Channel Assumptions

The design of FSO constellations is governed by unique physical constraints distinct from radio-frequency (RF) communications. First, Intensity Modulation/Direct Detection (IM/DD) systems impose a strict non-negativity constraint $x > 0$ on the transmitted symbols, as optical intensity represents power and cannot be negative [1]. Second, regarding temporal dynamics, we adopt the 'slow fading' assumption justified by Farid and Hranilovic [1], based on the fact that atmospheric coherence time ($\sim$ ms) is orders of magnitude longer than the symbol duration ($\sim$ ns). Consequently, the channel state h is treated as constant over a large block of symbols. Finally, consistent with standard capacity analysis [1], we assume the receiver has perfect Channel State Information (CSI) regarding the instantaneous fading realization h , while the transmitter optimizes the constellation based solely on the channel's statistical distribution.

Constellation Shaping Methods

To improve achievable information rates and reduce the shaping gap, constellation shaping techniques can be employed. Qu and Djordjevic [2] review two primary paradigms: Probabilistic Shaping (PS), which adjusts the transmission probabilities of fixed constellation points, and Geometric Shaping (GS), which maintains equiprobable symbols while optimizing point locations. Standard uniform constellations inherently suffer from an asymptotic shaping loss of $\pi \frac{e}{6} \approx 1.53 \text{ dB}$ compared to the Shannon limit [2]. Similarly, in FSO channels subject to signal-dependent noise and fading, standard constellations remain suboptimal. A comparative analysis by Qu and Djordjevic [2] indicates that the performance of shaping methods depends on the constellation size. Their findings demonstrate that Geometric Shaping (GS) achieves superior spectral efficiency compared to Probabilistic Shaping (PS) for lower modulation orders, while exhibiting comparable performance for larger constellations. Furthermore, the authors note that

GS avoids the implementation complexity associated with the distribution matchers (e.g., CCDFM) required for PS [2].

Optimization Objective Function

Defining the optimization metric is critical. Minimizing Symbol Error Rate (SER) is effective for uncoded systems but does not necessarily maximize information throughput. Pragmatic AMI (PAMI) is suitable for bit-interleaved coded modulation (BICM) but depends on labeling choices. Average Mutual Information (AMI) provides an information-theoretic measure of the achievable rate for a given discrete constellation under equiprobable inputs [2]. Therefore, AMI is selected as the objective function, aiming to maximize the achievable information rate under a fixed average transmit power constraint.

Optimization Algorithm - Simulated Annealing

The AMI-based constellation optimization problem is typically non-convex and may contain multiple local maxima. Gradient-based methods, which rely on local slope information, can become trapped in suboptimal solutions. To mitigate this, Simulated Annealing (SA) is employed as a probabilistic metaheuristic that explores the solution space broadly. Kayhan and Montorsi [3] demonstrated the effectiveness of SA for constellation design in channels with non-Gaussian likelihood characteristics, such as memoryless phase noise channels. Although their work focuses on phase noise, their methodology for using SA to maximize AMI in non-Gaussian channels is directly applicable to our log-normal intensity fading channel.

Furthermore, [3] highlights two critical practical aspects adopted in this work: First, regarding initialization, they propose a hybrid strategy where optimization begins from a standard constellation (e.g., Uniform-PAM) rather than random points, ensuring stable convergence. Second, regarding robustness, their studies indicate that optimized constellations exhibit superior resilience to channel parameter mismatches compared to rigid standard schemes, maintaining performance even when estimated channel conditions deviate from actual values.

In order to validate our understanding and the implementation of the Simulated Annealing algorithm presented in [3], we have successfully reproduced the constellation optimization results reported by the authors. A comparison between our simulation results and the original figures from [3] is provided in Appendix A.

Proposed Solution

Based on the survey of different approaches, we propose to solve the problem using Geometric Shaping (GS) optimized via the Simulated Annealing (SA) algorithm, with the objective of maximizing the Average Mutual Information (AMI). The design considerations leading to this selection are as follows:

GS vs. PS: Geometric Shaping (GS) was selected based on Qu and Djordjevic [2], who demonstrated that GS yields superior spectral efficiency compared to Probabilistic Shaping (PS) for small constellation sizes, while achieving equivalent performance for large constellations. Therefore, for the modulation orders considered in this project, GS optimizes channel capacity effectively without the additional hardware complexity required for PS.

SA vs. Gradient Descent: The AMI landscape for this channel is likely complex and non-convex. SA is chosen for its ability to escape local optima and find a potentially better global optimum compared to gradient-based methods, which might get stuck in suboptimal solutions.

AMI vs. SER/PAMI: AMI is selected as the objective because it represents the fundamental theoretical capacity limit for the given constellation and uniform input distribution, providing a clear benchmark for performance improvement. While SER is relevant practically, optimizing AMI targets the overall information rate. PAMI is relevant for coded systems but adds complexity (labeling optimization) not essential at this initial stage.

Proposed Implementation Strategy

Our solution implements a constellation optimization engine based on Geometric Shaping (GS). The core process involves iteratively adjusting the locations of M constellation points to maximize the AMI objective function using Simulated Annealing (SA). To balance computational efficiency with accuracy, we employ a multi-tier strategy:

1. Optimization Engine (Simulated Annealing):

We utilize a geometric cooling schedule ($T_{k+1} = \alpha T_k$) to gradually reduce the "temperature," allowing the algorithm to escape local optima in early stages and converge to a global maximum in later stages. The proposal step size is adaptive, dynamically adjusting based on the acceptance rate. At each iteration, the average optical power constraint is strictly enforced by normalizing the constellation vector.

2. Fast Objective Evaluation (Inner Loop):

Inside the optimization loop, where thousands of evaluations are required, we estimate the AMI using a Monte Carlo method combined with Gauss-Hermite quadrature (MC+GH). By applying the variable change $t = \ln(h)$, we approximate the integral over the log-normal fading efficiently, significantly reducing runtime while maintaining sufficient accuracy for the stochastic search.

3. High-Precision Validation (Outer Loop):

Once an optimal constellation candidate is found, its performance is verified using an independent Grid-Based Monte Carlo method (MC-via-grid). This method utilizes a dense, non-uniform grid to numerically integrate the likelihoods without the approximations used in the GH method, ensuring the reported Shaping Gain is accurate.

4. Parallel Initialization Strategy:

To ensure optimization robustness and maximize computational efficiency, we employ a parallelized hybrid initialization strategy. We will simultaneously execute independent optimization threads starting from both Uniform-PAM and randomly generated constellations, leveraging multi-core processing capabilities. The final solution will be selected from the thread achieving the highest validated AMI, ensuring that the algorithm reaches a global optimum and is not trapped in a local maximum.

Constraints

- Average Power Constraint: The optimization is performed under a fixed average transmit power constraint $P_{avg} = \frac{1}{M} \sum_{i=1}^M x_i^2$.
- Positive Intensity: Since the system employs Intensity Modulation (IM/DD), the transmitted optical signal must be strictly non-negative $x_i \geq 0$.
- Signal-to-Noise Ratio (SNR): The optimization is conducted for a specific, given SNR value (defined by the ratio of average signal power to noise variance), which is treated as a fixed parameter during the optimization run.

System Model Assumptions

- Weak Turbulence Regime: The atmospheric turbulence is assumed to be in the weak regime, modeled by a Log-Normal distribution for the fading intensity h . This is the standard model for clear-weather FSO links over moderate distances [1].
- Slow Fading: Since the atmospheric coherence time is orders of magnitude longer than the symbol duration, we assume a Slow Fading model. The channel coefficient h is considered constant over a transmission block of N symbols and changes independently between blocks.
- Channel State Information (CSI)
- Perfect receiver CSI: We assume the receiver has perfect knowledge of the instantaneous channel realization h (via pilot symbols or estimation).
- Statistical transmitter CSI: The transmitter has no knowledge of the instantaneous h due to feedback latency, but knows the channel's statistical parameters (specifically, the turbulence strength σ_X^2).

- AWGN Noise: The dominant noise source at the receiver (thermal and shot noise) is modeled as signal-independent Additive White Gaussian Noise (AWGN) with variance $\frac{N_0}{2}$.
- Perfect Synchronization: We assume perfect time and frequency synchronization between the transmitter and receiver, focusing the research solely on the modulation gain.

Initial Risks Analysis

- Convergence to Local Optima: The optimization landscape of the AMI function is inherently non-convex and complex. There is a risk that the Simulated Annealing algorithm may get trapped in suboptimal local peaks instead of finding the global maximum. If this occurs, the resulting constellation will not realize the full potential of geometric shaping, leading to lower-than-expected performance gains.
- High Computational Complexity: The evaluation of the AMI objective function requires numerical integration over the log-normal channel statistics. Performing this integration thousands of times within the optimization loop creates a significant computational bottleneck. Excessive runtimes could limit the number of channel scenarios or modulation orders (M) that can be thoroughly investigated within the project timeframe.
- Insignificant Shaping Gain: There is a possibility that for the specific channel conditions (log-normal fading) and low SNR ranges investigated, the theoretical gain of Geometric Shaping over standard Uniform-PAM may be quantitatively small (e.g., negligible dB improvement). This would impact the perceived success of the method for this specific application.
- Sensitivity to Channel Mismatch: The optimization relies on a specific statistical model of the channel (defined by σ_x^2). A potential risk is that the optimized constellation may be overly sensitive ("overfitted") to these specific parameters. In a scenario where the actual channel conditions deviate slightly from the model, the optimized constellation might perform worse than a robust, standard Uniform-PAM scheme.

Project Scope

The project scope focuses on the development, implementation, and analysis of an optimization algorithm for finding optimal one-dimensional (PAM) constellations for an FSO channel under weak turbulence conditions and verifying the performance through simulation.

Project Deliverables:

- Optimization Algorithm (Source Code): Full implementation of the Simulated Annealing (SA) algorithm in a MATLAB environment, tailored to maximize the AMI objective function under power constraints.
- Channel Model and Simulation Definition: A complete software implementation definition of the considered channel model, including log-normal fading parameterization (weak turbulence), AWGN noise model, and all design constraints (nonnegativity, ordering, and average transmit-power constraint).
- Constellation Database: Data files containing the optimal position vectors for a variety of SNR scenarios and turbulence strengths.
- Comparative Performance Report: A graphical and quantitative analysis comparing the performance of the geometrically shaped (GS) constellations against standard uniform constellations (Uniform PAM) and theoretical bounds, specifically quantifying the achievable Shaping Gain (in dB).

System Interfaces:

- Input: Channel parameters (Noise variance, Turbulence intensity), Average power constraint, and Constellation size (M).
- Output: The optimal location vector and the maximum AMI value achieved.

Project Boundaries:

- No hardware implementation: The project does not include building or testing a physical FSO transmitter/receiver, optical alignment hardware, or lab measurement.
- No full coded-system design: The project does not include full FEC/BICM system implementation, labeling optimization, or maximizing PAMI as a primary objective.
- Limited channel model: Unless explicitly extended, the scope does not include detailed pointing-error modeling, strong-turbulence models (Gamma–Gamma), hardware nonlinearities/saturation, or temporal correlation/memory effects.

Proposed Final Test Framework

The final acceptance testing aims to validate the correctness of the optimization algorithm and to quantify the performance gain achieved by the geometrically shaped constellations. Since this is a research project, the testing procedure focuses on verifying simulation accuracy and demonstrating improvement over the baseline.

Product Description:

The tested "product" consists of the optimal constellation coordinates vector generated by the Simulated Annealing (SA) algorithm for a given set of channel parameters, and the corresponding calculated AMI value.

Test Methodology:

The validation process will consist of two main phases:

- **Phase A:**

Algorithm Validation:

Test: Run the algorithm on a standard AWGN channel (without fading) where optimal solutions are known or can be easily computed numerically.

Measurement: Compare the resulting constellation and AMI against known literature results for AWGN capacity.

- **Phase B:**

Performance Verification on FSO Channel:

Test: Generate optimized constellations for Log-Normal fading channels with varying turbulence strengths.

Measurement: Calculate the AMI of the optimized constellation using an independent, high-precision numerical integration method (MC-via-grid with a dense grid, distinct from the fast estimator used in the optimization loop). This ensures the optimization did not exploit approximation errors.

Acceptance Criteria:

The project will be considered successful if the following criteria are met:

- **Shaping Gain:** The optimized GS constellation must achieve a higher AMI than the standard Uniform-PAM constellation for the same average power constraint. Specifically, we target a gain equivalent to at least 0.2 dB in SNR for high turbulence scenarios.
- **Convergence Consistency:** The SA algorithm demonstrate a consistent optimization trajectory across multiple independent runs with different random initializations. While exact final values may vary due to the stochastic nature of the algorithm, all runs should exhibit a similar improving trend and stabilize at a comparable high-quality solution.

Measurement Tools and Accuracy:

- Tools: The tests will be performed using a dedicated MATLAB simulation environment.
- Accuracy: To ensure statistical significance, Monte Carlo simulations for final verification will use at least $N = 10^5$ samples, providing a confidence interval sufficient to distinguish the Shaping Gain from numerical noise.

Budget estimation

This project is entirely theoretical and research-oriented, focusing on algorithm development and simulation. It does not involve the procurement of hardware components, manufacturing, or external services. All necessary computational resources and software licenses (MATLAB) are provided by the academic institution's existing infrastructure. Consequently, there are no direct financial costs associated with the project's execution, and the estimated budget is 0\$.

Schedule and division of labor

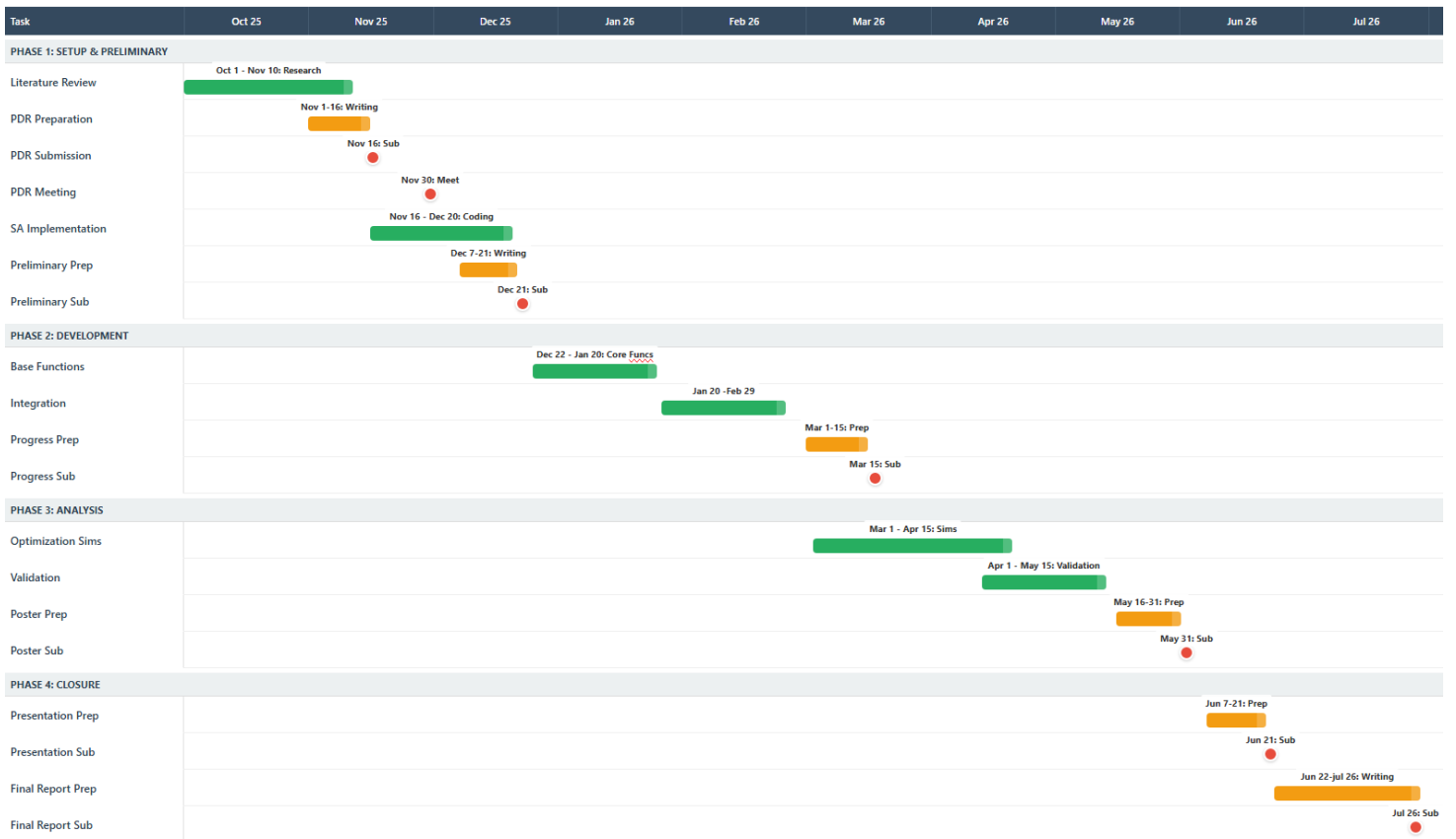
The project execution plan is designed to ensure the timely completion of all research and development objectives in alignment with the faculty's official submission deadlines. The timeline covers the period from the initial research phase (October 2025) through to the final report submission (July 2026).

Roles and Responsibilities:

While specific technical responsibilities are divided to maximize efficiency, the team maintains close collaboration and continuous mutual consultation throughout all development stages. Furthermore, team members are committed to providing mutual backup and support to address any scheduling constraints and ensure project continuity.

- Dolev Ishay: Focuses on the Optimization Engine. Responsibilities include implementing the Simulated Annealing framework, managing the optimization loop parameters, executing large-scale simulation runs for various scenarios, and extracting raw optimization data.
- Adi Shlomo: Focuses on the Channel Model & Verification. Responsibilities include implementing the physical channel model functions (Log-normal PDF, MC+GH integration), developing the independent verification method (MC-via-grid) to validate results, and performing the quantitative performance analysis.
- Joint Tasks: Literature review, system integration, debugging, preparation of presentations (Progress, Conference), and writing of the Preliminary and Final reports.

GANTT



References

- [1] A.A. Farid and S. Hranilovic, “Outage Capacity Optimization for Free-Space Optical Links With Pointing Errors”, J. Lightwave Technol. 25, 1702-1710 (2007).
- [2] Z. Qu and I.B. Djordjevic, “On the Probabilistic Shaping and Geometric Shaping in Optical Communication Systems”, IEEE Access 7, 21454 (2019).
- [3] F. Kayhan and G. Montorsi “Constellation Design for Memoryless Phase Noise Channels”, IEEE T. Wirel. Commun. 13, 2874 (2014).

Appendices

Appendix A: Optimization Algorithm Validation

A.1 Introduction and Objective:

As part of the preliminary design phase, and to verify the correctness of our Simulated Annealing (SA) algorithm implementation, we performed a replication of existing results from the professional literature. The purpose of this appendix is to present the comparison between the simulation results developed by us and the results published in reference [3] (Kayhan & Montorsi, "Constellation Design for Memoryless Phase Noise Channels").

This validation serves to prove that the optimization engine, the cooling schedule, and the acceptance criterion (Metropolis Criterion) function correctly and converge to the optimal solution, prior to their application to the specific FSO channel model of this project.

A.2 Experimental Setup:

The validation simulations were conducted using the MATLAB environment developed for this project. We configured the system with the following parameters, matching those in the reference article[3]:

- Modulation Order: $M = 8$.
- Power Constraint: Average power normalized to unity ($P_{avg} = 1$).
- Channel Model: Additive White Gaussian Noise (AWGN) combined with Phase Noise modeled by a Tikhonov distribution.
- Optimization Target: Maximization of Average Mutual Information (AMI).

A.3 Result I:

Optimal Constellation Geometries First, we examined the geometric structure of the constellations obtained at a fixed SNR of 12 dB under varying phase noise intensities (PNSD).

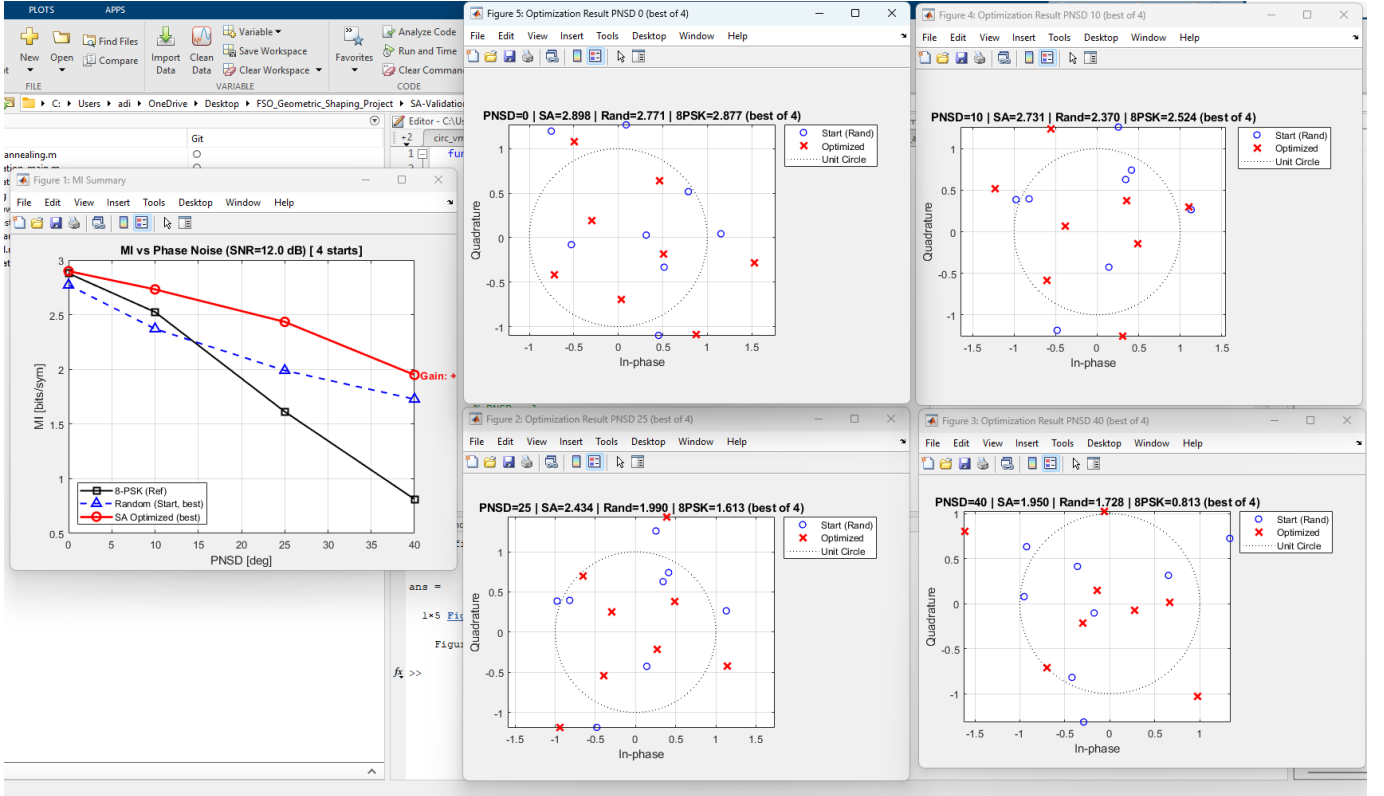


Figure A.1: Our results for optimized constellation geometries for M=8 at SNR=12dB with varying Phase Noise (PNSD). The red crosses represent the optimized locations, while blue circles represent the random initialization.

Reference results:

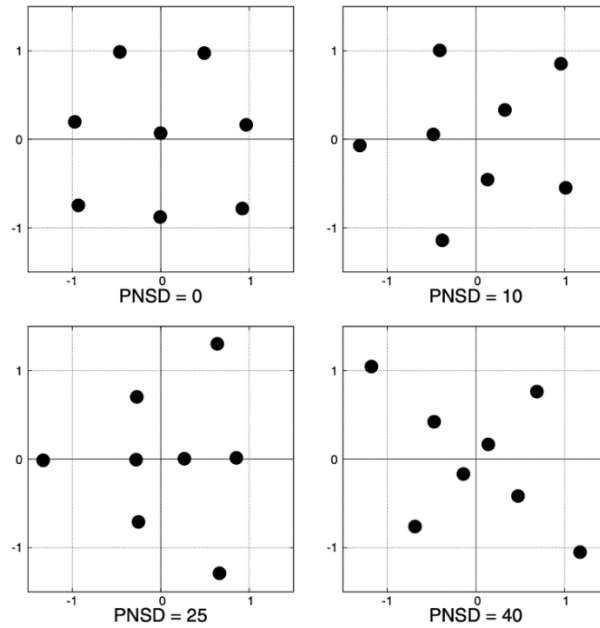


Figure A.2: Reference results from [3] (Fig. 2. Optimized 8-ary constellations for four different values of PNSD. The objective function is the AMI and the SNR is fixed into 12 dB.)

Validation Results: As demonstrated in the simulation results (Figure A.1), our algorithm successfully reproduces the key performance trends described in the literature [3]:

- **Shaping Gain Analysis:** Consistent with [3], the results show that at low phase noise levels (e.g., $\text{PNSD} = 0^\circ$), the shaping gain is minimal as the channel behaves similarly to an AWGN channel. However, as the phase noise increases (e.g., $\text{PNSD} = 40^\circ$), the Shaping Gain becomes significant, with the optimized constellation drastically outperforming the standard 8-PSK.
- **Geometric Similarity:** The optimized constellations obtained by our algorithm exhibit geometric structures similar to those reported in [3].

A.4 Result II:

AMI Performance vs. SNR to quantify the shaping gain, we expanded the simulation to sweep across a range of SNR values (1 dB to 13 dB) for different phase noise levels. Our results of this performance analysis are presented in Figure A.3.

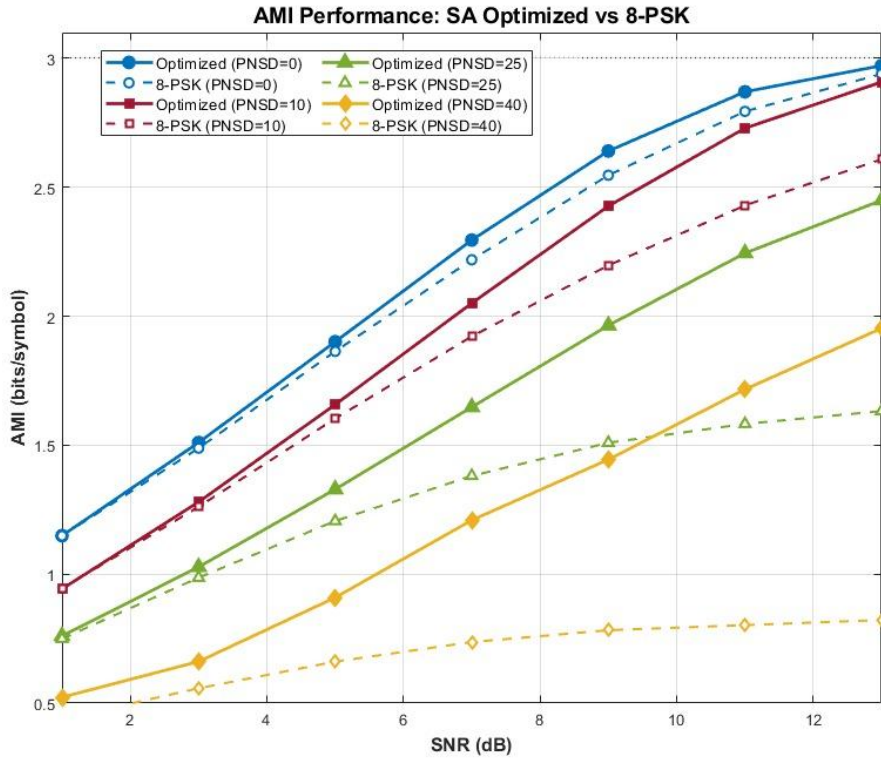


Figure A.3: AMI performance comparison between the SA-Optimized constellations (Solid lines) and standard 8-PSK (Dashed lines) across various SNR and PNDS levels.

Reference results:

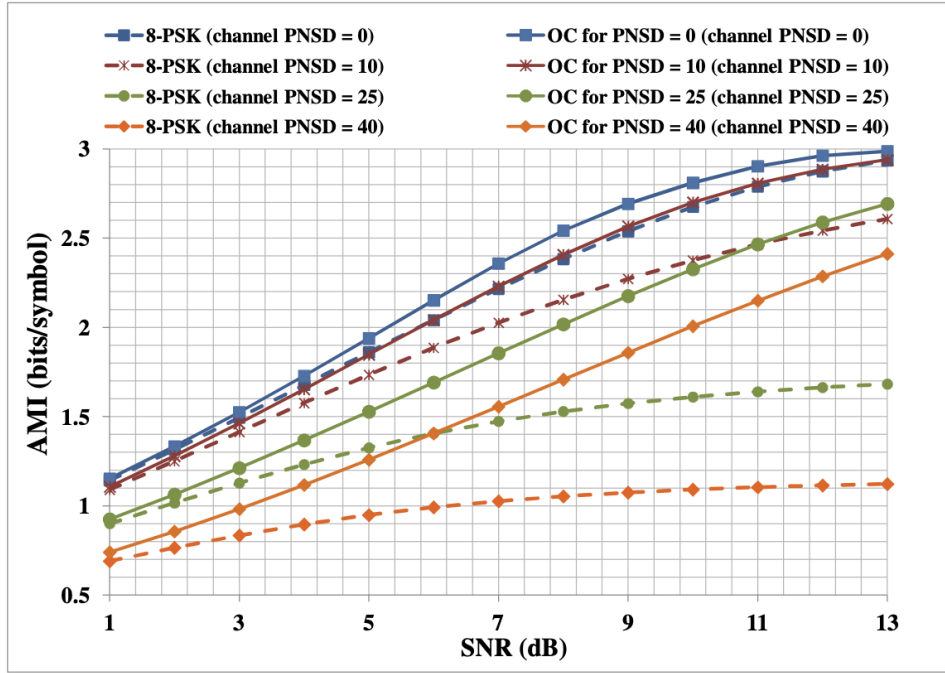


Figure A.4: Reference results from [3] (Fig. 5. Performance of the 8-PSK constellation as a function of SNR for various PNSD values. The solid lines are the AMI for the optimized constellations. The dashed lines are the AMI for the 8-PSK used over the channel with the indicated PNSD value.)

AMI Performance Trends: Figure A.3 illustrates the AMI performance comparison, demonstrating trends that closely align with the reference literature [3]. The results indicate that at low SNR and low phase noise levels (e.g., $\text{PNSD}=0^\circ$), the Shaping Gain is minimal, as the standard 8-PSK constellation provides near-optimal performance. However, a significant divergence is observed at high SNR and high phase noise levels (e.g., $\text{PNSD}=40^\circ$), where the optimized constellation achieves a substantial Shaping Gain. This confirms that the Geometric Shaping algorithm effectively mitigates capacity loss in severe noise regimes.

A.5 Conclusions:

The successful replication of both the geometric structures (Fig A.1) and the AMI performance curves (Fig A.3) validates our optimization framework. The Simulated Annealing algorithm demonstrated the ability to escape local minima and converge to solutions that provide measurable Shaping Gain. Based on these results, we are confident in applying this engine to the Log-Normal FSO channel in the next phase of the project.

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תאריך: 21/12/2025

חתומים: דולב ישי (211721451) עדי שלמה (211619226)

Grade recommendation for a preparatory report:

מספר הפרויקט: P-2026-093 שם הפרויקט: Geometric shaping in the FSO turbulent channel.
 שם המנחה פרופ' דרביאנקו סטניסלב.
 שמות הסטודנטים: ישי דולב, שלמה עדי.

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